# AUTUMN MID SEMESTER EXAMINATION-2025

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University

**DATA MINING AND DATA WAREHOUSING**
**CS30013**

**Time: 1 1/2 Hours** **Full Mark: 20**

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***Answer any four questions including question No.1 which is compulsory.***
***The figures in the margin indicate full marks.***
***Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.***

1. Answer all the questions. [ 1 Mark X 5 ]

a. Obtain the trimmed 20% mean for the following data: 25, 28, 30, 32, 35, 36, 40, 42, 45, 48, 67, 78, 87, 89, 90.

b. How continuous valued attribute is mapped into discrete value, discuss any one technique with a suitable example?

c. What is the role of metadata in managing and querying a multidimensional schema?

d. How do feature selection and dimensionality reduction differ in their impact on knowledge discovery?

e. Given, Support ({Butter, Bread}) = 0.3, Support({Butter}) = 0.4, Support({Bread})= 0.5, Find the lift of the rule Butter → Bread.

2. A class has scores summarized as follows: [10,15,16,18,21,25,26,27,31,51,55,53,55,29,61]

   a. Is the data symmetric? [5 Marks ]
   b. Identify any potential outliers.
   c. Describe the spread.
   d. Draw the boxplot analysis.

3. Consider the following 3D data cube for a retail store with dimensions Product (P1, P2), Region (North, South), and Time (Q1, Q2). Each cell stores the Total Sales (in ₹ '000). [5 Marks ]

| Product | Region | Q1 | Q2 |
|---|---|---|---|
| P1 | North | 40 | 60 |
| P1 | South | 30 | 50 |
| P2 | North | 20 | 40 |
| P2 | South | 10 | 30 |

Using roll-up, slice and dice operations on this cube, answer the following:

a. Roll-up the cube on the Time dimension to show total sales for each (Product, Region) pair.

b. Slice the cube to show sales of Product P1 only across Region and Time.

c. Dice the cube to select data where Region = North and Time = Q2, and report the result.

4. Apply the Pearson Correlation Coefficient and the Cosine Similarity method to find the relationship between the following two variables. [5 Marks ]

| Student | Study Hours (X) | Exam Score (Y) |
|---|---|---|
| A | 2 | 65 |
| B | 3 | 70 |
| C | 5 | 80 |
| D | 7 | 90 |

5. Find the frequent item-sets for the given database (D) by using Apriori algorithm by considering a minimum support of 40% and generate strong association rules using minimum confidence of 70%. Calculate the lift for each strong association rule to evaluate the correlation between the antecedent and the consequent.

[ 5 Marks ]

| TID | List of Items |
|---|---|
| 1 | I1, I2, I3 |
| 2 | I3, I4 |
| 3 | I4, I5 |
| 4 | I1, I2, I5 |
| 5 | I1, I2, I4 |
| 6 | I2, I3, I4 |
| 7 | I2, I3 |

*** Best of Luck ***